

Profit & Loss, Discount

**By:-
Naveen Kumar Jain
(Aptitude Trainer)**



PROFIT, LOSS & DISCOUNT

Cost Price (CP)

The rate at which a merchant buys goods. This is his investment

Selling Price (SP)

The rate at which a merchant sells his goods.

Marked Price (MP)

The rate at which a merchant rises his price above the cost price (may be anticipating some hagglers)

Profit Case

If Selling Price is more than the Cost Price

$$[SP > CP]$$

Loss Case

If Cost Price is more than the Selling Price

$$[CP > SP]$$

Key Formulas:

1. Profit or Gain = Selling Price - Cost Price = $SP - CP$
2. Profit % = $(\text{Profit}/CP) \times 100 \% = (SP - CP)/CP \times 100 \%$
3. Loss = Cost price - Selling price = $CP - SP$
4. Loss % = $(\text{Loss}/CP) \times 100 \% = (CP - SP)/CP \times 100 \%$

Important: Profit or Loss always calculated on Cost price only.

5. Discount = Marked price - Selling Price = $MP - SP$
6. Discount % = $(\text{Discount}/MP) \times 100 \% = (MP - SP)/MP \times 100\%$
7. If two products are sold at the same selling price, one with a profit of $x \%$ and another with a loss of $x \%$, then the net Loss % = $x^2/100 \%$

1.

(A) If C.P = 125, S.P = 96, then, Loss = ?

(B) If C.P = 112, S.P = 132, then, Gain = ?

(C) If C.P = 120, S.P = 90, then, Loss% = ?

(D) If C.P = 80, S.P = 100, then, Gain% = ?

(E) If C.P = 90, Gain% = 10%, then, S.P = ?

(F) If C.P = 20, Loss% = 25%, then, S.P = ?

(G) If S.P = 84, Gain% = 20%, then, C.P = ?



2. A watch costing Rs 250 has been sold for Rs. 300. The Percentage profit was:
- A. 20 B. 15 C. 16.66 D. 25

**3. A shopkeeper sells almonds at the rate Rs 114 a kg and bears a loss of 5%.
If he sells it at Rs 126 per kg, what will be the result?**

- A. 15% gain B. 5% loss C. 5% gain D. 15% loss**

4, Mr. Mirza Baig on selling a Furniture for Rs 450 bears a loss of 20%. To earn a profit of 20%, he should sell the article for:

A. Rs. 675

B. Rs. 600

C. Rs. 625

D. Rs. 680

5. A Man sells an article at a loss of 10%. If he had received Rs. 9 more, he would have gained 12.5%. The cost price of the article is:

A. Rs. 40

B. Rs. 36

C. Rs. 32

D. Rs. 48



6. A person marks his goods 20% higher than the cost price and allows a discount of 5%. The percentage profit is:

- A. 15 B. 20 C. 5 D. 14**

7. What is a single discount equivalent to a discount series of 10% and 20% ?
- A. 25 B. 30 C. 28 D. 15

8. A dishonest shopkeeper professes to sell pulses at the cost price, but he uses a false weight of 950gm. for a kg. His gain is ____%.

- A. 5% B. 11% C. $4\frac{4}{19}\%$ D. $5\frac{5}{19}\%$

According to Q , He sells 950 gm of pulses And gets a gain of 50 gm

$$\begin{aligned}\text{Gain}\% &= \frac{50}{950} \times 100 \\ &= 5\frac{5}{19}\%\end{aligned}$$

9. A shopkeeper increased the price of a product by 50% and later on he reduced the price by 50%. Shopkeeper's loss was:

- A. 0% B. 2.5% C. 25% D. 0.25%**

If the cost price of m objects is equal to the selling price of n objects, then the net

$$\text{Profit/Loss \%} = (m - n)/n \times 100 \%$$

If Cost price of 6 article is equal to selling price of 5 article .find profit or loss %?

Cost price of 6 article = Selling price of 5 article

Here m=6 and n=5

$$\text{Profit/Loss \%} = (m - n)/n \times 100 \%$$

$$\text{Profit \%} = (6 - 5)/5 \times 100 \%$$

$$\text{Profit \%} = 1/5 \times 100 \%$$

$$\text{Profit \%} = 20\%$$

10. The cost price of 20 pencils is equal to the selling price of 25 pencils. The loss percent in the transaction is:

A. 5

B. 20

C. 25

D. 30

By selling 15 notebooks a merchant gains the selling price of 3 notebooks. Find its gain %?

Given gain of 3 notebooks, when we sell SP of 15 notebooks

SP of 15 notebooks = CP of 15 notebooks + SP of 3 notebooks

SP of 12 notebooks = CP of 15 notebooks

$$\begin{aligned}\text{Gain \%} &= \frac{15-12}{12} \times 100 \\ &= 25\%\end{aligned}$$

11. A dishonest milkman buys milk at Rs. 6 per liter and add one third of water to it and sells the mixture at Rs. 7.20 per liter. The gain is:

A. 40%

B. $80/3\%$

C. 60%

D. 25%



12. A retailer buys 40 pens at market price of 36 pens from a whole seller. If he sells these pens giving a discount of 1%. What is profit % ?

A. 20%

B. 12.5%

C. 10%

D. 11.11%

- 13. The percentage increase in the number of pens sold by a trader from Tuesday to Wednesday is same as the percentage decrease in the number of pens sold from Monday to Tuesday. On Wednesday the trader sold 16 pens more than those he sold on Tuesday. If the percentage decrease in sales from Monday to Tuesday is 20%, how many pens did he sell on Tuesday?**
- A. 20 B. 96 C. 100 D. 80**
- 

14. Two men, sitting on the dining table. One man has 7 eggs and other had 5 eggs. A third man passing by requested them to share their food in return for money. The three of them shared the eggs equally and the third traveler paid the other two a total of Rs 24. Find the difference between the amounts received by first two men?

A. Rs. 18

B. Rs. 12

C. Rs. 16

D. Rs. 15



15. A sold a watch to B at a gain of 5%, and B sold it to C at a gain of 4%. If C paid Rs. 1092 for it, the price paid by A was

A. Rs. 990

B. Rs. 1200

C. Rs. 1000

D. Rs. 1100



16. I sold 2 watches for Rs. 300 each, one at the loss of 10% and the other at the profit of 10%. What is the percentage of loss or profit resulted from the transaction ?
A. 10% profit B. 1% loss C. 1% profit D. 10% loss



17. By selling 12 articles, a shopkeeper gains a profit equal to the cost price of 2 articles. What is his profit percentage ?

A. 16.66%

B. 17.14%

C. 18.28%

D. 20.33%

18. Use the data given in the table to fill the empty cells.

Cost Price	Selling Price	Profit	Loss	Profit%/Loss%
Rs. 580			Rs. 120	
Rs. 1250		Rs. 1400		
4 for Rs. 7	7 for Rs. 10			
Rs. 560				24% Profit

19. A hat vendor bought hats at Rs. 5 per 3 hats. He sold them at Rs. 10 per 4 hats. What was his profit percentage?

A. 25 B. 30 C. 45 D. 50



20. A dishonest businessman professes to sell his articles at cost price but uses false weights to cheat the buyers by 10% while buying, and by 10% while selling. Finds the profit percentage.

A. 11.11%

B. 15.00%

C. 22.22%

D. 50.00%



21. Alan Sells a bag at 25% profit. Had he bought it for 20% less and sold it for Rs. 10.50 less, he would have gained 30%. Find the cost price of the bag.

A. Rs. 50 B. Rs. 100 C. Rs. 150 D. Rs. 200



22. A retailer offers two successive discounts of 15% and 25% on a shirt marked at Rs. 1600. How much will a customer have to pay for the shirt ?

A. Rs. 960 B. Rs. 1000 C. Rs. 1020 D. Rs. 1120



**23. A merchant decides to sell his goods at C.P but wants to get a 25% profit.
How many grams should he give instead of a kg?**

A. 750 B. 800 C. 850 D. 900

24. A man bought two dozen apples. He sold half of them at the cost price of 10 apples and the remaining half at the cost price of 15 apples. What is the overall gain or loss percent ?

- A. 6.25% loss B. 6.33% gain C. 6.67% loss D. 4.16% gain**
- 

25. A merchant decides to sell his goods at C.P. but also wants to make 25% profit. How many grams should he give instead of a kg ?
A. 750 B. 800 C. 850 D. 900



26. If 4 cars are sold at the cost price of 6 cars the profit % will be
A. 12.5% B. 16.67% C. 33.33% D. 50.00%

27. A man sold two articles at Rs. 990 each. He gained 10% on the first article and bore a loss of 10% on the second. What is his approximate gain or loss?

A. Rs. 0 B. Rs. 10 C. Rs. 20 D. Rs. 32



28. Sam buys rice at Rs. 10/kg and puts a price tag on it so as to earn a profit of 20%. However, his faulty balance shows 1000 gm when it is actually 800 gm. What is his actual gain percentage?

A. 50%

B. 40%

C. 28%

D. 20%



29. A shopkeeper sells pencils at a profit of 20% for Rs. 60. But due to the fall in demand he reduces the price to Rs. 55. Find the percentage of gain or loss he now makes.

A. 10%

B. 20%

C. 30%

D. None of these



30. Instead of listing an article at 20% above its cost price and selling it at a discount of 25%, a trader listed it at 25% above its cost price and sold it at a discount of 20%. What is the gain or loss to the trader because of the change in selling price in terms of original selling price?

A. 10% loss

B. No loss, no gain

C. 10% gain

D. 11.11% gain



31. If two successive discounts of 10% and 20% are allowed on the marked price of Rs.500 and still a profit of 9.09% is obtained, what is the cost price?

A. Rs.300 B. Rs.330 C. Rs.360 D. Rs.400



32. A merchant gives 40% discount on an article and still gains 20%. How much percentage above the cost price is the article marked?

A. 30%

B. 40%

C. 50%

D. 100%



33. A shopkeeper professes to sell all things at a discount of 25% but marks each article 20% above the cost price. What is his gain or loss percentage?

- A. 5% loss B. 10% loss C. 10% gain D. 15% gain**

34. Use the data given in the table to fill the empty cells.

Cost Price	Marked Price	Selling Price	Discount %	Profit/Loss %
180	220	200		
335	350		35%	
140		100	40%	

35. A reduction of 33.33% in the price of oranges would enable a purchaser to obtain 8 more for a rupee. What was the price in paise, per orange, before the reduction ?

A. 5 paise

B. 6.25 paise

C. 6 paise

D. 5.25 paise



- 36. What is the percent loss on cost price if the loss on selling price is Rs. 20?**
- A. 16.66% B. 18.23% C. 21.33% D. 5.66%**

37. If a bike is sold for Rs. 30,000 then a man loses 25%. At what price must that bike be sold in order to gain 25%?

- A. Rs. 40,000 B. Rs. 50,000 C. Rs. 55,000 D. Rs. 60,000**

38. The gain for shopkeeper in certain grocery item is 30%. If the total sale of an item in a day was Rs. 550. Find the profit of shopkeeper.

A. 110 B. 127 C. 138 D. 149



39. A man sold his second hand mobile phone at Rs. 7500 by losing one-sixth of the price at which he bought. Find the cost of the mobile phone.

A. Rs. 8200

B. Rs. 8300

C. Rs. 8700

D. Rs. 9000



40. A shopkeeper claims to sell a pipe at cost price but, being dishonest, he sells the pipe of 90cm instead of a meter. Find the profit of dishonest shopkeeper.

A. 9% B. 11.11% C. 19.43% D. 24.74%



THANK YOU

